

MAT 1272 Statistics - Quiz 6

Name: _____

Instructions: Show all work. Clearly identify the hypotheses, test statistic, p-value, calculator command, and conclusion.

Question (20 points)

A consumer organization claims that the average battery life of a certain laptop is at least 8 hours. A random sample of 25 laptops is tested and produces a mean battery life of 7.4 hours. Assume the population standard deviation is 1.8 hours. Using a 5% significance level, test the claim.

- (a) State the null and alternative hypotheses.
- (b) Which hypothesis represents the claim?
- (c) Is the test left-tailed, right-tailed, or two-tailed?
- (d) Will you use a z-test or a t-test? What is the critical value?
- (e) Find the p-value. Clearly show the TI-84 command used.
- (f) State your decision and conclusion in one complete sentence.